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Regulatory Post-Market Surveillance Quality Systems 510(k) US

FDA Surgical Mesh Recalls: 171 Actions and the Polymeric-Mesh MAUDE Record

Surgical mesh analysis: 171 FDA recalls, 15 Class II enforcements, and 145,000 MAUDE reports. Bard/Davol leads recalls and reports; the POP-mesh 510(k) line was pulled to Class III in 2016.



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Published 2026-06-15 Last reviewed 2026-06-15 12 min read



Executive Summary

Surgical mesh is the device family that exposed the limits of the 510(k) predicate chain. Our analysis of the complete FDA recall, enforcement, 510(k), PMA, and MAUDE record for the surgical-mesh product-code family finds a category whose regulatory pathway split in two

- **171 FDA recall events** match the surgical-mesh product-code family, and **15 text-matched enforcement reports** are recorded across the category; the enforcement record is overwhelmingly Class II, with the historic C.R. Bard Kugel patch the notable Class I action

- **Davol, Inc. (C.R. Bard, now BD) dominates the recall record** with 21 mesh-family recalls, followed by AROA Biosurgery (21), Boston Scientific (14), Cook Biotech (13), and TEI Biosciences (11); process control, device design, and labeling are the leading root-cause categories

- **The MAUDE adverse-event record is one of the largest in surgery**: we count **145,310 device-event reports under the surgical-mesh product-code family across 2015–2026, including 1,259 death reports and 139,452 injury reports**, with annual volume spiking to 26,229 reports in 2019 amid the transvaginal-mesh litigation

- **The pathway split is the defining fact**: 780 510(k) clearances and only five PMA rows exist under the mesh family codes — because most mesh is still Class II 510(k), while the two urogynecologic pelvic-organ-prolapse codes (OTP, PAI) were reclassified to Class III in 2016 and ordered off the U.S. market entirely in April 2019

For quality, regulatory, and surgical-supply-chain teams, the takeaway is specific: hernia and abdominal-wall mesh remains a 510(k) Class II device where the post-market burden concentrates in manufacturing process control and packaging integrity at a small group of incumbent manufacturers, while the urogynecologic POP-mesh segment that triggered the regulatory crisis is effectively closed to new 510(k) entry.

Why Surgical Mesh Became a Regulatory Test Case

Synthetic surgical mesh entered U.S. commerce through the 510(k) pathway using a small set of predicates. The FDA cleared the first mesh product for stress urinary incontinence — Boston Scientific's ProteGen Sling — in 1996 and recalled it roughly three years later, but

not before several additional products had been cleared against its design (Drugwatch). Manufacturers ultimately brought at least 61 mesh implants to market through 510(k) by leveraging predicates such as Ethicon's Mersilene Mesh and Boston Scientific's ProteGen Sling, even where the predicate had itself been recalled (Heneghan et al., *F1000Research*).

The post-market data eventually forced a regulatory split. In 2014, citing a sharp rise in adverse-event reports for mesh used in transvaginal pelvic-organ-prolapse (POP) repair, the FDA issued two orders proposing to reclassify synthetic and non-synthetic mesh for transvaginal POP repair from Class II to Class III. The orders were finalized in 2016, requiring manufacturers to submit PMA applications by July 5, 2018 for any mesh intended for transvaginal POP repair (TorHoerman Law).

On April 16, 2019, after reviewing the PMA submissions, the FDA ordered all remaining manufacturers of surgical mesh for transvaginal POP repair — Boston Scientific (Uphold Lite, Xenform) and Coloplast (Restorelle DirectFix) — to stop selling and distributing their products immediately, determining they did not demonstrate reasonable assurance of safety and effectiveness under the Class III standard (FDA, 2019). No PMA was ever granted; the entire transvaginal POP-mesh segment was withdrawn from the U.S. market.

This analysis covers the surgical-mesh product-code family — the codes under 21 CFR 878.3300 (polymeric, collagen, and metal surgical mesh) and 884.4910/884.5980 (urogynecologic mesh):

Representative product codes	Device	FDA class
FTL	Mesh, Surgical, Polymeric (general hernia / abdominal wall)	II
JDJ / OXB / OWV	Mesh, Surgical — acetabular / collagen thoracic / collagen diaphragmatic hernia	II
OTP	Mesh, Surgical, Synthetic, Urogynecologic, for Pelvic Organ Prolapse	III (reclassified 2016; withdrawn 2019)
PAI	Mesh, Surgical, Non-Synthetic, Urogynecologic, for Pelvic Organ Prolapse	III (reclassified 2016; withdrawn 2019)

Forty-three product codes fall in this family: 41 Class II and 2 Class III. The Class II codes cover the full hernia, abdominal-wall, and soft-tissue reconstruction range that remains on the market; the two Class III codes are the transvaginal POP-mesh segment that the FDA closed.

Data Source and Method

- **Sources**: FDA Medical Device Recall database (recall initiation records by product code), FDA device enforcement report database (Class I/II/III classification and reason-for-recall text), FDA 510(k) and PMA databases, and the FDA MAUDE adverse-event database
- **Analysis sample**: 171 recall events under the 43 mesh-family product codes; 15 enforcement-report records whose product description and reason-for-recall text matches surgical-mesh terminology; 780 510(k) clearances and 5 PMA records under the same codes; 145,310 MAUDE device-event reports under the mesh-family codes across 2015–2026
- **Run date**: 2026-06-15
- **Method**: All counts were computed by MedDeviceGuide from the public FDA recall, enforcement, 510(k), PMA, and MAUDE extracts. Enforcement records do not carry a populated product-code field, so mesh records were identified by text matching of the device description and reason-for-recall fields; firm names are reproduced as recorded by FDA. MAUDE event-type counts (death, injury, malfunction) reflect reports filed, not confirmed causal attributions, and a single adverse-event report can contain multiple device-event rows.

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The Recall Footprint

Across the 171 surgical-mesh recall events under the family product codes, the recalling-firm concentration is sharp:

Recalling firm (as recorded by FDA)	Mesh-family recall events
Davol, Inc., Sub. C. R. Bard, Inc.	21
AROA Biosurgery, LTD.	21
Boston Scientific Corporation	14
Cook Biotech, Inc.	13
TEI Biosciences, Inc.	11
Synovis Surgical Innovations (Div. of Baxter)	10

Aggregated across the longer history, manufacturers have recalled **more than 211,000 hernia mesh products since 2005** (ConsumerNotice.org). The recall-status distribution of the 171 family events shows an active long tail: 117 Terminated, 46 Open/Classified, and 8 Completed. Root-cause themes tallied from the recall records separate manufacturing-quality failures from device-design and labeling failures:

Root-cause category	Recall events
Other / unclassified	51
Under investigation by firm	24
Process control	19
Device design	13
Incorrect or no expiration date	12
Nonconforming material / component	10

Process control (19), labeling and expiration integrity (12), and nonconforming material (10) together account for the largest actionable share — these are manufacturing and packaging-process problems addressable through standard 21 CFR 820 process controls, not intrinsic device-design flaws.

The Enforcement Record: Mostly Class II, One Historic Class I

Text-matching the FDA enforcement database for surgical-mesh devices returns **15 enforcement reports, all classified Class II** over the period covered — the classification that signals temporary or medically reversible adverse health consequences. The one widely cited Class I action in the mesh category sits outside the current family-code window: C.R. Bard's Composix Kugel patch, recalled for "memory recoil ring" breakage that could cause bowel perforation and chronic enteric fistula, was the rare mesh action classified at the highest severity (Drugwatch; ConsumerNotice.org).

The current enforcement picture is otherwise a Class II manufacturing and packaging record — a useful corrective to the public perception that mesh recalls are uniformly life-threatening. The genuinely serious mesh harm, as the MAUDE data shows, is a chronic-complication and reoperation problem recorded as injury reports rather than a recall-classification problem.

The MAUDE Adverse-Event Record

The MAUDE adverse-event database is where the mesh safety story is most visible. Under the 43 mesh-family product codes, we count **145,310 device-event reports across 2015–2026, including 1,259 death reports, 139,452 injury reports, and 4,455 malfunction reports**.

Annual report volume shows a clear 2018–2019 surge, coinciding with the transvaginal-mesh litigation, the FDA's 2018 PMA deadline, and the April 2019 market-withdrawal order, followed by a decline as POP-mesh reporting normalized and the remaining hernia-mesh installed base drove a steadier baseline:

Year (date received)	MAUDE mesh-family reports
2015	9,387
2016	6,919
2017	5,849
2018	13,954
2019	26,229
2020	21,177
2021	20,948
2022	13,555
2023	10,094
2024	10,418
2025	4,860
2026 (partial)	1,920

The FDA separately reports that it had identified **approximately 55,000 adverse-event reports related to hernia mesh** as of a July 2023 update — a hernia-specific figure that triangulates the scale of the hernia-mesh segment within the broader family count above (FDA, Surgical Mesh for Hernia Repair). The 2018 surge is visible at the monthly level too: the FDA averaged more than 500 surgical-mesh adverse-event reports per month that year, including roughly 300 for Ethicon Physiomesh and 200 for Atrium C-QUR in the first quarter alone (Drugwatch).

By manufacturer, the MAUDE footprint mirrors the recall footprint. Davol, Inc. (C.R. Bard, now BD) carries the largest share at roughly 61,000 reports (including both "Davol Inc., Sub. C.R. Bard, Inc." and related entity filings), followed by Ethicon Inc. (28,964), Sofradim Production (20,181, the Covidien/Medtronic mesh manufacturing entity), Boston Scientific (5,990), and Atrium Medical (4,217):

Brand name (as filed in MAUDE)	Reports
BARD FLAT MESH	7,650
PERFIX PLUG	7,563
PARIETEX	6,244
VENTRALIGHT ST MESH	5,781
VENTRALEX ST	5,309
GYNECARE MESH	4,553
3DMAX	3,934
PROGRIP	3,758
PROCEED SURGICAL MESH	3,200

The clinical problem mix — pain, adhesion, hernia recurrence, seroma, and disability-injury-pain clusters — is the chronic-complication signature that drove more than **100,000 transvaginal-mesh lawsuits** in the POP segment and, as of June 2026, approximately **26,153 hernia-mesh lawsuits across four federal multidistrict litigations** (C.R. Bard/Davol MDL 2846 with ~23,573 pending cases, Covidien MDL 3029 with ~2,408, Atrium C-QUR MDL 2753 with ~172, and Ethicon Physiomesh MDL 2782 largely resolved) (ConsumerNotice.org; Motley Rice).

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The 510(k) Entry Record and the Predicate Legacy

Despite the post-market burden, the hernia and abdominal-wall mesh segment remains open to new 510(k) clearances. We count **780 510(k) clearances under the mesh-family**

product codes, with the applicant field concentrated in the same incumbent firms:

Applicant (as filed)	510(k) clearances
W. L. Gore & Associates, Inc.	48
C.R. Bard, Inc.	45
Sofradim Production	41
ETHICON, Inc.	40
American Medical Systems, Inc.	34
Cook Biotech, Inc.	21

Clearance activity has been steady in recent years — 13 in 2021, 13 in 2022, 15 in 2023, 14 in 2024, 22 in 2025, and 4 in the first half of 2026 — reflecting a mature, consolidated market with a stable incumbent set and modest new entrant flow.

The predicate-ancestry problem is what made the mesh category a regulatory test case. Academic analysis of the mesh predicate network identified three recalled meshes — Composix Kugel Mesh (K003323), PROCEED Trilaminate SurgicalMesh (K031925), and PROCEED Surgical Mesh (K060713) — that had been on the market for up to five years before recall and that served as predicates for subsequently cleared devices, raising structural questions about whether a recalled predicate should propagate new clearances (Heneghan et al., *F1000Research*). The PROCEED delamination recalls (Ethicon recalled more than 18,000 units of PROCEED Surgical Mesh from 2005 to 2014 over packaging-related delamination) illustrate the chronic packaging-integrity failure mode in the hernia-mesh Class II record (Drugwatch).

The two Class III urogynecologic POP-mesh codes (OTP, PAI) have effectively no live 510(k) activity: after the 2016 reclassification and the April 2019 withdrawal order, that segment is closed to new 510(k) entry and requires a PMA that no manufacturer has obtained.

What This Means for Quality, Regulatory, and Supply-Chain Teams

The data supports four practical takeaways:

1. **Treat hernia mesh as a process-control and packaging problem, not a design-classification problem.** The recall root-cause record is dominated by process control, expiration/labeling, and nonconforming-material categories — all addressable through 21 CFR 820 manufacturing controls. The segment remains Class II 510(k), and the post-market burden is an execution risk at incumbent manufacturers.

2. **Read the MAUDE record as a chronic-complication signal, not an acute-failure signal.** Mesh harm is overwhelmingly recorded as injury (pain, adhesion, recurrence, seroma) rather than malfunction; benchmarking against malfunction counts understates the clinical risk that drives both litigation and reoperation.

3. **Recognize the closed POP-mesh segment.** The OTP and PAI codes were reclassified to Class III in 2016 and withdrawn in 2019. Any residual transvaginal POP-mesh product strategy is non-viable in the U.S. without a successful PMA, and none exists.

4. **Manage supplier concentration as litigation concentration.** Because Bard/Davol, Ethicon, Sofradim/Covidien, Atrium, and Cook carry both the largest MAUDE footprints and the largest MDL exposures, a health system's effective mesh recall and litigation exposure is concentrated in three or four supplier relationships.

Surgical mesh is the device family whose post-market record forced the FDA to split a single 510(k) category into a still-open Class II hernia segment and a closed Class III urogynecologic segment. The public data — 171 recalls, 145,310 MAUDE reports, and a 780-clearance 510(k) record anchored by recalled predicates — is the evidence base that reclassification was built on, and it remains the benchmark for current mesh safety performance.

Data sources: FDA Medical Device Recall database, FDA device enforcement report database, FDA 510(k) and PMA databases, and FDA MAUDE adverse-event database; analysis by MedDeviceGuide, run date 2026-06-15. Root-cause themes are derived from FDA recall and reason-for-recall text and are not mutually exclusive. MAUDE event-type

counts reflect reports filed, not confirmed causal attributions. This article is educational and is not regulatory, quality-system, legal, or clinical advice for a specific product.

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